

Examiner reports

Q1.

This was a very well answered question overall. Parts (a), (b), (c) and (d) were all answered correctly by the majority of students, with many making very good use of the calculator. Only part (e) proved problematic, with many students failing to score, often as a result of making comments not clearly enough related to the binomial conditions, or otherwise by not stating them sufficiently in context.

Q2.

Overall students performed quite well on this question, but there were a significant minority who appeared insufficiently prepared for this topic and unable to gain much credit. In general, students performed well on parts (a) and (b), with the majority scoring full marks. Most students did not score well on part (c), failing to appreciate the significance of likely negative values within the 3 standard deviation spread. Part (d) had a very mixed response. Many students scored full marks but many others could not use the inverse normal distribution correctly.

Q5.

Whilst full marks were very occasionally seen to this question, scores of 4 or less were by far the most common. Most students had little or no idea as to how to answer this question. In answering part (a), common incorrect answers were 10, $\sqrt{10}$ (3.16), 20 or 100 whereas correct answers of 5 ($10 / 2$) or preferably 3.33 ($10 / 3$) were rarely seen. Some students scored 1 mark in part (b) for giving a reasonably clear argument as to why 391 was likely to be incorrect. However, even when part (a) was answered correctly, a valid reason as to why 95.5 was likely to be incorrect proved a step too far for most students. In part (c), most students scored 1 mark for $8210 / 10 = 821$ or for $10 \times 820 = 8200$. A straight comparison of 821 with 820 scored a second mark, but those students who found 8200 were required to indicate that 8210 was within 100, not 10, for this second mark. It was very disappointing in part (c) to see those of students who failed to realise that it was necessary to calculate $\sqrt{\frac{110}{9}}$ or $\sqrt{\frac{110}{10}}$ in order to comment on the value of $\sum(y - \bar{y})^2$, particularly given the formula for s^2 in the booklet. All too often, they argued that $\sqrt{110} \approx 10$ which, in turn, approximately equalled their (incorrect) value of σ in part (a).

Q6.

Most candidates scored well on this question involving normal distributions, with full marks often achieved in part (a). However, weaker candidates lost 2 marks in part (a)(ii) for calculating $P(X < 25)$ as 0.242. Most such candidates then lost a further 2 marks in part (a)(iii) by evaluating [(i) – (ii)] since, whilst this gave the correct answer of 0.546, it was the result of two errors. All but the weakest candidates were able to make a good, often fully correct attempt at

part (b), usually by working with $\frac{\frac{65}{1.34} - 32}{10}$. Some lost an accuracy mark for rounding $\frac{65}{1.34}$ to 48 or 49, whilst others lost 2 marks for evaluating $P(\text{bill} < 65)$. A minority of

candidates attempted to work with the distribution of the bill and, whilst they had the correct mean of 42.88, they left the standard deviation as 10. Answers to part (c) were poor and rarely scored the 2 marks available. It was expected, from the wording of the question, that candidates would recognise the possibility of other types of customer, fuel and / or vehicle. However, most candidates either concentrated on the actual distribution of X or commented on the variation in unleaded petrol sales due to factors already accounted for by the distribution $N(32, 10^2)$.

Q7.

This question proved a good source of marks for almost all candidates; clearly they had a very sound grasp of the applications of binomial distributions. By far the most common approach in part (a) was to use the formula in part (a)(i) and then tables in parts (a)(ii) and (a)(iii). In part (a)(iii), the usual confusions sometimes arose as to whether to use the values for $x = 9$ or 10 or for $x = 4, 5$ or 6 . In part (b), the only noticeable error was to evaluate $(32 \times 0.15 \times 0.85)$ as 4.08 but then not take the square root; a loss of 1 mark here. Almost all candidates calculated correctly, using their calculators' mean and standard deviation functions, the mean and the standard deviation of the given 13 values. Many then went on to compare the 2 mean values and the 2 standard deviation values and so comment on the proposed model's suitability. Centres are advised that comments of the form 'means and standard deviations are different' is not sufficiently clear for the awarding of marks.

Q8.

The changes in values of p through this question proved to be quite challenging for some students. However, only the weakest students failed to score some marks in part (a)(i);

often due to the omission of $\binom{30}{2}$ or an error in the indices. Part (a)(ii) caused more students a problem with a significant number trying to perform separate calculations for $p = 0.22$ and $p = 0.13$ before giving up since probabilities for neither value were tabulated. Those students who worked with $p = 0.35$ usually then made correct use of tables, rather than multiple formula calculations, although extracting a probability for an incorrect value (9 instead of 10) was still too prevalent. Disappointingly, given its regular examination on previous papers, for these and other students this uncertainty continued into part (a)(iii) which often resulted in the loss of at least one mark. Almost all students obtained correct values for the mean and variance in part (b)(i) although, for some students, there was some uncertainty as to whether the value of 17.16 was the variance or the standard deviation. Answers to part (b)(ii) were very disappointing with few students scoring more than one mark. All too often students compared the two means and the two standard deviations (using $\sqrt{17.6}$) but then did not even comment on the pairwise similarities or, more often than not, they simply stated 'results valid' or 'accept both claims'. In order to improve responses to such requests, teachers may wish to view the mark scheme to see what was required for the 3 marks.

Q9.

This was probably the best answered question on the paper with the better candidates often scoring the full 14 marks available. Apart from the small minority of candidates who found, by formulae, $P(R = 7)$ and $P(R = 10)$ in parts (a)(i) and (a)(ii) respectively, almost all candidates attempted to use the cumulative binomial tables in these parts. In part (a)(i),

$P(R = 7)$ was usually found correctly but, in part (a)(ii), some candidates used $P(R \leq 11)$, instead of $P(R \leq 10)$, whilst others forgot to subtract their values obtained from the tables from 1. Part (a)(iii) caused some candidates more difficulty, with many uncertain as to how to determine $P(5 < R < 10)$. Whilst most candidates subtracted two cumulative probabilities, one or both were often incorrect values and, given that this type of question has appeared regularly on previous papers, calculations such as $P(R \leq - (1 - P(R \leq 5)))$ were particularly disappointing.

Most answers to part (b), based on using the binomial formula, were correct, although a small minority of candidates used $n = 14$ or 42 or even $n = 20$, presumably so that tables could be used. Determination of the mean and variance in part (c)(i) was carried out correctly by almost all candidates. However, many candidates struggled to express their comparisons of results using appropriate language. Phrases involving “correct”, “incorrect” or “exact” were not appropriate, and neither was the word “different” when comparing 1.43 to 1.50; rather the two values were “similar” or “about/nearly the same”.

Q10.

Most candidates scored at least 8 of the 10 marks available in part (a) by making appropriate use of the binomial formulae for a probability, mean and variance and for using cumulative binomial tables. Where marks were lost, it was usually for finding $P(X \leq 15)$ in part (a)(ii) and/or for subtracting $P(X \leq 15)$ from $P(X \leq 20)$ in part (a)(iii).

Only a very small number of candidates were able to give a satisfactory reason, such as the likely non-independence of senior citizens’ travel, in part (b)(i). However most candidates gained the mark available in part (b)(ii), where a variety of acceptable reasons were offered.

Q11.

Candidates appeared better prepared for part (a) than in the past, so their answers, usually directly from calculators, were in the main correct. It was also pleasing to note the large proportion of correct answers to parts (b)(i) and (ii). In the former, some candidates mysteriously obtained the correct answer despite totally incorrect work in part (a) and, in the latter, some answers were backed-up by a correct calculation.

In part (b)(iii), only the most able candidates were able to make valid statements as to why a normal distribution was not an appropriate model. Many invalid attempts made reference to the sample size or the Central Limit Theorem, neither of which had any relevance here. Candidates should be aware that, if X is normally distributed, then, for any sample size, the sample mean, $\bar{X}$, will be normally distributed regardless of the Central Limit Theorem; for other cases, the Central Limit Theorem states that, for sufficiently large sample size, the sample mean will be approximately normally distributed, but candidates should not infer from this that the population values, or actual sample values, will themselves be approximately normally distributed.

Q12.

Candidates appeared better prepared for part (a) than in the past but it remains a concern that far too many candidates cannot sensibly tackle this type of question. Whilst most candidates were able to correctly identify the median, the IQR appeared to be an unknown term or was found using $75 - 25 = 50 \Rightarrow \text{median} = 3$. A small minority of

candidates even based their answers on the cumulative values of $\sum fx$. – In part (a)(ii), answers, usually directly from calculators, were in the main correct, although incorrect answers due to working only with x -values or only with f -values were seen.

It was pleasing to note the large proportion of correct answers to parts (b)(i) and (ii). In the former, some candidates mysteriously obtained the correct answer despite totally incorrect work in part (a) and, in the latter, some answers were backed-up by a correct calculation.

In part (b)(iii), only the most able candidates were able to make valid statements as to why a normal distribution was not an appropriate model. Many invalid attempts made reference to the sample size or the Central Limit Theorem, neither of which had any relevance here. Candidates should be aware that, if X is normally distributed, then, for any sample size, the sample mean, $\bar{X}$, will be normally distributed regardless of the Central Limit Theorem; for other cases, the Central Limit Theorem states that, for sufficiently large sample size, the sample mean will be approximately normally distributed, but candidates should not infer from this that the population values, or actual sample values, will themselves be approximately normally distributed.

Q13.

Virtually all candidates used the appropriate binomial tables but frequently made one or more errors by finding $P(R \leq 10)$ in part (a)(i) and/or using $P(R \leq 9)$ or $P(R \leq 5)$ in part (a)(ii).

Candidates appeared well prepared in the use of the binomial formula, with almost all scoring full marks in part (b)(i). However, answers to part (b)(ii) were very disappointing. All too often, candidates calculated $P(S = 1)$ or $1 - P(S = 1)$ rather than $P(S \geq 1) = 1 - P(S = 0)$, with some even using $n = 22$ rather than 35.

In answering part (b)(iii), most candidates took note of the word 'correctly' and the number '120' in the question to then score both marks. However, a minority lost marks for stating

answers of 7.2 or $2.60(\sqrt{6.768})$. Part (b)(iv) proved beyond all but the high achievers. Whilst some candidates based their comments on the perceived practical difficulties in sorting letters, others decided that there was only one assumption and so made their comments unclear. Those candidates who did treat the statements as two separate assumptions, still often demonstrated little understanding of the implications of 'equal means' but 'different variances'. Many such candidates used the means to comment on independence and the variances to comment on the probability. As a result, the awarding of full marks for this final part was very rare.

Q14.

Parts of this question proved a good source of marks for many candidates with the most able scoring all 12 marks. Answers to part (a) were usually correct and found, as was intended, from tables. A minority of candidates calculated $P(M = 15)$ using the formula. Most candidates used the correct formula in part (b), but a small minority approximated 0.29 by 0.30 so as to use tables, resulting in a loss of all of the 3 marks available. Answers to part (c)(i) showed an improvement in the knowledge of the relevant formulae. Only a very small proportion of candidates apparently chose to ignore the emboldened word '**do**' and used $p = 0.29$ rather than $p = 0.71$, resulting in a loss of at least 4 marks since the comment marks in part (c)(ii) were dependent upon correct answers in part (c)(i). Answers to part (c)(ii) were somewhat disappointing. Many candidates made no

reference to their (correct) answers in part (c)(i) but simply stated that “the samples could not be random as they only included women”. Only better candidates were able to score at least 2 of the 3 marks available. The mark lost was usually for stating that “since the samples were not random, the claim was not justified”.

Q15.

Parts of this question proved a good source of marks for many candidates with the more able scoring all 15 marks. Answers to part (a)(i) were usually correct and found, as was intended, from tables. A minority of candidates calculated $P(M = 15)$ using the formula. As always, part (a)(ii) caused difficulties to many candidates. Whilst almost all of them attempted a subtraction using values from tables, many considered at least one incorrect value of M . The requirement of ‘more than 10 but fewer than 20’ translated to $P(10 < M < 20) = P(M \leq 19) - P(M \leq 10)$. Most candidates used the correct formula in part (b), but a significant minority approximated 0.29 by 0.30 so as to use tables, resulting in a loss of all of the 3 marks available. Answers to part (c)(i) showed an improvement in the knowledge of the relevant formulae. A small proportion of candidates apparently chose to ignore the emboldened word ‘do’ and used $p = 0.29$ rather than $p = 0.71$, resulting in a loss of at least 4 marks since the comment marks in part (c)(ii) were dependent upon correct answers in part (c)(i). Nevertheless, many candidates were able to score at least 2 of the final 3 marks. When a mark was lost, it was usually for stating that “since the samples were not random, the claim was not justified”. A number of candidates made no references to their answers in part (c)(i), but simply stated that “the samples could not be random as they only included women”.

Q16.

Many candidates scored at least 10 marks on this question. Almost all of them showed a correct expression in part (a) and then evaluated it correctly. Answers to part (b)(i) were also almost always correct with just a small minority of candidates evaluating $P(\text{Continental} = 12)$ from the formula. In part (b)(ii), there was the usual confusion between ‘ $\leq$ ’ and ‘ $<$ ’ when using Table 1 but the majority of candidates were able to score at least 2 marks. In part (c), a minority of candidates did not use $p = 0.7$, whilst many more either misread the request or were under the impression that $np(1 - p)$ gave the standard deviation.

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Q17.

This proved to be a good source of marks for many candidates with the more able often scoring full marks. Answers to part (a)(i) were usually correct from using Table 1 in the booklet provided. The minority using the formula often calculated $P(T = 3)$. Part (a)(ii) caused some difficulties. Whilst almost all candidates attempted a subtraction of values from Table 1, many had difficulty in translating ‘on more than 10 days but fewer than 20 days’ to $P(R \leq 19) - P(R \leq 10)$ with most errors involving the use of $P(R \leq 20)$ and/or $P(R \leq 9)$.

Knowledge of the relevant formulae needed to answer part (b)(i) was much improved. Failure to take the square root of 1.2 was the usual reason for forfeiting marks. As expected, part (b)(ii) was answered correctly by the vast majority of candidates. Answers to part (b)(iii) frequently contained comments that were valid, consistent and appropriate.

Q18.

Most candidates had a sound knowledge of calculating binomial probabilities. Many fewer candidates appeared to know how to calculate the mean and variance of a binomial distribution, whilst some appeared not to have studied the binomial distribution. A large majority of candidates scored full marks in parts (a)(i) & (ii). In part (a)(iii), confusion often arose as to whether the tabulated values required were for 7 or 8 and for 4 or 3. As a result many candidates lost some marks. A small minority of candidates opted for using the binomial formula sometimes with complete success. In part (b), those candidates with the necessary knowledge scored both marks. Others sometimes obtained the mean value by intuition but of course could make no headway with finding the variance. Yet again, far too many candidates appear unaware of the information provided in the first table on page 11 of the supplied booklet.

Q19.

Most candidates scored full marks in parts (a)(i) & (ii); the latter usually from the binomial formula. Answers to part (a)(iii) were less impressive. Whilst many candidates had basically the correct idea, attempts of the form $P(R \leq 15) - P(R > 5)$ or $P(R > 5) - P(R \leq 14)$ were not uncommon and a very few candidates attempted repeated use of the formula. Answers to part (b) were again disappointing. Many answers were based solely on Sly's calculated values. When the value for the binomial mean was stated, it was often as a result of intuition rather than the formula. On the somewhat rare occasions when the value for the binomial variance was found, comparisons of means and variances were done in isolation and an overall conclusion was lacking for the final mark.

Q20.

This question proved a good source of marks for many candidates with the more able often gaining at least 11 marks. Answers to part (a)(i) were often correct as most

candidates realised that $7.5 \left(\frac{15}{2} \right)$ required $P(K \leq 1)$ or $P(K < 8)$ and so scored full marks from tables. A small minority of candidates used $P(K = 7)$ or $\frac{P(K \leq 7) + P(K \leq 8)}{2}$.

As in previous papers, part (a)(ii) continued to cause more difficulties almost always through use of $P(K \leq 3)$ and/or $P(K < 7)$. Knowledge of (an available source of) the relevant formulae for part (b)(i) was much improved with many fully correct answers as a

result. When marks were lost, it was invariably for failing to find $\sqrt{3.6}$ for the standard deviation. Not surprisingly, only the very weakest candidates failed to score the 2 marks in part (b)(ii). In part (b)(iii), able and well-prepared candidates compared the two means (equal) and two standard deviations (different) but then often failed to commit to the overall conclusion of 'doubtful validity'. A small minority of candidates only compared

$\frac{\bar{x}}{15} = \frac{6}{15} = 0.4$ with the value quoted (1 mark) or presented a qualitative argument as to the benefits of coaching (0 marks).

Q21.

This question proved a good source of marks for many candidates with the more able often gaining at least 15 marks. Answers to part (a)(i) were almost always correct and

usually by use of the formula. In part (a)(ii), most candidates realised that $7.5 \left(\frac{15}{2} \right)$ required $P(K \leq 7)$ or $P(K < 8)$ and so scored full marks from tables. A small minority of

candidates used $P(K = 7)$ or $\frac{P(K \leq 7) + P(K \leq 8)}{2}$. As in previous papers, part (a)(iii) continued to cause more difficulties almost always through use of $P(K \leq 3)$ and/or $P(K < 7)$. Knowledge of (an available source of) the relevant formulae for part (b)(i) was much improved with many fully correct answers as a result. When marks were lost, it was invariably for failing to find $\sqrt{3.6}$ for the standard deviation. Not surprisingly, only the very weakest candidates failed to score the 2 marks in part (b)(ii). In part (b)(iii), able and well-prepared candidates compared the two means (equal) and two standard deviations (different) but then often failed to commit to the overall conclusion of 'doubtful validity'. A minority of candidates only compared $\frac{\bar{x}}{15} = \frac{6}{15} = 0.4$ with the value quoted (1 15 15 mark) or presented a qualitative argument as to the benefits of coaching (0 marks).

Q22.

Parts (a) and (b) were considered to be routine tasks and it was disappointing to see that some candidates appeared unable to make any attempt.

Answers to part (a)(i) were usually correct, although a small minority used $p = 0.04$, rather than 0.4, in the cumulative binomial tables. Part (a)(ii), caused candidates all sorts of difficulties. Whilst most candidates tried to use the aforementioned tables, many could not deal correctly with both inequalities. Thus $P(X \leq 5)$ was often used instead of $P(X \leq 4)$ and, although many candidates found $P(X \leq 1)$ correctly, they then used this to find $P(X > 1)$ and subsequently $P(X \leq 5) - P(X > 1)$ as their final answer. The alternative method of using binomial formulae usually involved correct expressions but many candidates failed to evaluate them correctly.

Again, in part (b), binomial expressions were generally correct, although some candidates used $n = 7$, instead of 28, whilst others gave an answer to less than three significant figures.

Surprisingly, many candidates omitted part (c), this despite the specific reference to the topic in the Specification and the provision of relevant formulae in the supplied booklet. Some candidates, using the formulae, gave the value of the variance rather than the required standard deviation. Other candidates were able to intuitively state the mean correctly without explicit knowledge of its formula.

In part (d)(i), some candidates ignored the frequencies. Otherwise, most candidates used the statistical functions on their calculators with most obtaining both the mean and standard deviation correctly, although a minority reversed values and frequencies. The few candidates who had a complete set of answers to parts (c) and (d)(i) for comparison, were generally reluctant to commit to a conclusion and were often confused in their interpretations. Some candidates who failed to answer part (c) essentially made use of mean = np by showing that their mean in part (d)(i) when divided by 7 gave the stated probability of 0.4.

Q23.

Most candidates made a confident start to this question. In part (a)(i), they usually standardised 60 using 56 and 2.5 though a very small minority used $\sqrt{2.5}$ or 2.5^2 in the

denominator. However rather more candidates standardised 59 (0 marks) or 59.5 (1 mark). Part (a)(ii) was also answered well with many candidates scoring full marks. When marks were lost, the reason was usually for not dealing with $\Phi(-2.4)$ as $1 - \Phi(2.4)$; a difficulty that may not have arisen had such candidates indicated the required area on a sketch of a normal curve.

Very few candidates had correct answers to part (a)(iii). Apparently few candidates were aware that $P(X = x) = 0$ when x is continuous and so often attempted $P(54.5 < X < 55.5)$ despite there been only 1 mark available. A majority of candidates scored at least 3 of the 4 marks available in part (b); this lost mark was through using $z = +2.0537$. A number of

candidates lost at least 3 marks for equating $\frac{100 - \mu}{3.4}$ to the probability for $z = 0.98$ from Table 3 of the supplied booklet.

Q24.

Statistics 1A

Candidates were often able to score at least some of the first 4 marks using the statistical functions on their calculators, particularly as the mark scheme allowed for some 'errors' in finding the class mid-points that were not integers. Candidates using formulae were often much less successful usually through working only with either class mid-points or class frequencies. When attempted by the more able candidates, answers to part (b) often referred to skewness or non-symmetry, although statements referring to the discrete nature of the data or likely negative values were equally valid. However statements on the size of the standard deviation were deemed invalid. Most seen answers to part (c)(i) were correct through mention of the Central Limit Theorem, or large sample size, although in many scripts it appeared that this was perhaps as good a place as any to state it. Answers to part (c)(ii) were much less impressive with typical common mistakes equivalent to mean $= \mu = 25.2$ and variance $= 17.2$ or 17.2^2 . Despite this, many candidates recovered to make a valid attempt at part (d) by using a correct expression with a correct z -value. Here the final accuracy mark also required accurate answers to part (a). Finally, in part (e), some candidates failed to notice that the first claim referred to 'average' and so 30 required a comparison with their confidence interval. To score the marks, this comparison had to be clear. A number of other candidates simply compared 30 with their sample mean. However, a surprising number of candidates answered the second claim correctly by identifying that only 7% of the sample values exceeded 50; in some cases these were the only marks scored on the question.

Statistics 1B

Candidates were often able to score the first 4 marks using the statistical functions on their calculators, particularly as the mark scheme allowed for some 'errors' in finding the class mid-points that were not integers. Candidates using formulae were often much less successful, usually through working only with either class mid-points or class frequencies. When attempted by the more able candidates, answers to part (b) often referred to skewness or non symmetry, although statements referring to the discrete nature of the data or likely negative values were equally valid. However statements on the size of the standard deviation were invalid. Most answers seen to part (c)(i) were correct through mention of the Central Limit Theorem or large sample size. Answers to part (c)(ii) were much less impressive with typical common mistakes equivalent to mean $= \mu = 25.2$ and variance $= 17.2$ or 17.2^2 . Despite this, many candidates recovered to make a valid attempt at part (d) by using a correct expression with a correct z -value. Here the final accuracy

mark also required accurate answers to part (a). Finally, in part (e), some candidates failed to notice that the first claim referred to 'average' and so 30 required a comparison with their confidence interval. A number of other candidates simply compared 30 with their sample mean. However, a pleasantly surprising number of candidates answered the second claim correctly by identifying that only 7% of the sample values exceeded 50. A minority of candidates compared 50 with their confidence interval.

Q25.

Notation and presentation of method were in need of improvement on many scripts although many candidates scored high marks for correct answers. In part (a), almost all candidates knew how to standardise 250 using 200 and 25 and so obtained a correct answer. A minority of candidates used 249, 249.5 or 249.9, employed a divisor of 625 or 5, or, having found $z = 1.8$ correctly, chose to use 1.79 in Table 3. Fewer candidates were completely successful in part (a)(ii), usually through not making the area change, $\Phi(-0.2) = 1 - \Phi(0.2)$. In part (b)(i), many candidates made the mistake of using 0.5244 or $1 - 0.5244$ rather than -0.5244 , whilst a minority used 0.70 as a z -value in Table 3. More candidates answered part (b)(ii) correctly since the z -value needed was positive, but a minority read 'medium' as 'median' and so used $z = 0$. There was a surprisingly large proportion of correct answers to part (c) often from candidates who had answered part (b) incorrectly. One general point of note for the future: too many candidates lose valuable accuracy marks through finding approximate z -values from Table 3 rather than more exact values from Table 4.

Q26.

Answers to parts (a)(i) & (ii), using the binomial formula and cumulative tables respectively, were in the main of a good standard although in some scripts it was surprising to see poor attempts at part (i) followed by correct attempts at part (ii). Part (b) generally proved beyond all but the most able candidates. The most common approach was to state $P(Y \geq 30) = 1 - P(Y \leq 29)$ and then simply switch to $p = 0.45$ in order to use Table 2. Some better candidates who reached $P(Y \leq 20 | p = 0.45)$ then changed the correct answer of 0.2862 to 0.7138. Weaker candidates also often made no attempt at part (c), this despite the fact that it offered relatively easy marks. Those candidates who answered part (i) correctly, sometimes stated the variance as their answer to part (ii) or used $n = 10$. Many candidates, even with full marks for parts (i) & (ii), simply accepted the model in part (ii) because of 'two outcomes' and 'fixed trials'. Rarely did candidates comment on the discrepancy between the two values for the standard deviation.

Q27.

This opening question allowed almost all candidates to score some marks since standardising was almost always of the correct form. In part (a) a significant number of candidates obtained the probability of 'less than 60 cm' and then found that they had an identical answer for part (b). Only a small proportion of such candidates corrected their answers to part (a) to zero. It was pleasing to see the number of candidates capable of scoring full marks in part (c). On the other hand, however, it was somewhat of a shock to see the number of weak candidates unable to find 5% of 48; sometimes this inability appeared disguised as 48 ± 0.05 or $48 \pm (z\text{-value})$.

Q28.

This question proved to be the best source of marks with the great majority of candidates scoring full marks. Candidates realised at last that in part (a), a description of the type of distribution (Geometric) and the value of the parameter ($p = 0.8$) were both required. Most of any errors seen in the remainder of the question were usually due to the candidates' inability to use their calculators correctly!

Q29.

As expected, almost all candidates answered part (a) correctly. However, in part (b), far too many candidates expressed $P(5 < X < 10)$ as one of $P(X = 5, 6, 7, 8, 9)$, $P(X \leq 10) - P(X \leq 5)$, $P(X \leq 10) - P(X \leq 6)$ or $P(X \leq 9) - P(X \leq 6)$ and so scored at most one mark. Given the competent handling of the normal approximation to the binomial on recent papers, it was somewhat disappointing to see the many candidates who either ignored the continuity correction or used 149.5, rather than 150.5, in part (c). Answers to part (d) were generally poor and rarely indicated that $p = 0.15$ would be unlikely for all vehicles or that the 50 successive vehicles may not be independent. The numerous suggestions that drivers of vehicles with faulty brake lights would not stop after observing the policewoman perhaps bodes ill for future accident statistics!

Q30.

This question was answered well by most candidates, who were able to score 5 out of 6 marks. Almost all candidates identified $B(20, 0.4 \text{ or } 0.6)$, though one or two used $p = 0.5$. In part (a), candidates tended to use the formula, usually with total success. However, some candidates spoilt their attempt by quoting 0.12, rather than 0.117, whilst others either doubled or squared a correct answer. In part (b), most candidates opted for using tables and then scored full marks. A minority of candidates chose to continue using the formula, some with success, but many often lost marks for omitting $P(O = 0)$ or for numerical errors. Answers using a normal approximation were not accepted. In part (c), very few candidates realised that the salient reason was that the number of trials was not fixed. Most candidates simply referred to a large sample or non-constant proportion.

Q31.

Nearly all candidates recognised the binomial distribution in part (a) but in part (a)(ii) many were unable to translate 20 or fewer invigorating cubes into 30 or more relaxing cubes.

There were many good answers to part (b) although some were able to list the necessary conditions for the binomial distribution but unable to apply them in the given context.

Q32.

This was not a high scoring question. Many candidates failed to realise that in part (a)(ii) they needed to calculate the probability of consumption being between 20 litres and 40 litres.

There were some good answers to part (b). Part (c) was not well answered. Although most candidates successfully found the z -values, the equations often contained incorrect signs which lead to nonsensical answers.

Q33.

Most candidates successfully identified $B(8, 0.3)$ but more than expected used the $p = 0.03$ column of the tables. There were a few excellent answers to parts (b) and (c) but, in the main, the explanations of why the binomial was not appropriate were rather shaky.

Q34.

Parts (a)(i), (a)(ii) and (c)(ii) were well answered. As usual, calculating the mean and standard deviation of a binomial distribution and of the observed data caused problems.

Q35.

Part (a) posed few problems but a wide variety of answers were given in part (b)(i). A number of candidates rounded the correct answer to part (b)(i) to 3 significant figures. In view of the rubric this was accepted but those who used this rounded figure in part (b)(ii) lost accuracy marks.

A few candidates used spurious “continuity corrections”. This should be discouraged, although it is usually treated sympathetically. On this occasion, weights were given to one decimal place and so using 124.5 (as several candidates did) instead of 125.0 is unarguably wrong. Disappointingly, there were few good answers.



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